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Dipartimento
di Fisica e Astronomia
"Galileo galilei" - DFA



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

DEPARTMENT
OF PHYSICS AND ASTRONOMY
"AUGUSTO RIGHI"



Admittance spectroscopy of organic-silicon heterojunction structures fabricated with ultrasound assistance

O. Olikh, D. Kalyuzhny, V. Kozachenko, S. Kondratenko

Taras Shevchenko National University of Kyiv, 64/13, Volodymyrska Street, 01601, Kyiv, Ukraine

olegolikh@knu.ua

Organic–silicon hybrid structures are promising for next-generation photovoltaics due to high carrier mobility and scalable fabrication. PEDOT:PSS/Si heterojunctions enable low-temperature processing and tunable interfacial energetics [1]. Because defects govern their electronic properties, their characterization and control are essential. Here, admittance spectroscopy [2] is employed to probe defect states, and acoustic waves are used for defect engineering.

The n-Si/PEDOT:PSS structures were fabricated by silicon substrate cleaning, spin-coating of the organic layer, room-temperature relaxation, annealing at 140 °C, and contact evaporation. Film thickness and interfacial quality were tuned by varying the spin-coating speed and relaxation time. Ultrasound treatment (UST) was applied during relaxation of some samples, using primary waves at 2.5 MHz or radial waves at 500 kHz.

The interfacial trap density of states (tDOS) was evaluated using the Walter formalism [2] based on capacitance–frequency measurements (100 Hz–1 MHz) at bias voltages of 0 and 0.4 V. The trap emission prefactor ($4.3 \pm 0.2 \cdot 10^4 \text{ s}^{-1} \text{K}^{-2}$) was extracted from temperature-dependent data in the range of 220–300 K (Fig. 1a). Defect states were resolved by Gaussian decomposition of the defect-related signal, with energy levels determined from peak positions and relative contributions quantified from area fractions (Fig. 1b).

Reducing the organic layer thickness by increasing the spin-coating speed slightly decreases the total defect density (Fig. 2). Forward bias increases the detected defect population due to enhanced contributions from deeper trap states (Figs. 2 and 3). UST has little effect on the absolute defect density but significantly reshapes the energy distribution by promoting deeper trap states, particularly under radial-wave excitation (Fig. 3). Although prolonged relaxation favors deeper levels, UST narrows the defect energy distribution. Overall, acoustic-wave-assisted fabrication enables effective defect engineering in organic–silicon hybrid structures.

[1] Z. Lu, J. Chen, Y. Wu *et al.*, *Appl. Phys. Lett.* **2025**; 127, 093901.

[2] T. Walter, R. Herberholz, C. Muller, and H. W. Schock, *J. Appl. Phys.* **1996**, 80, 4411.

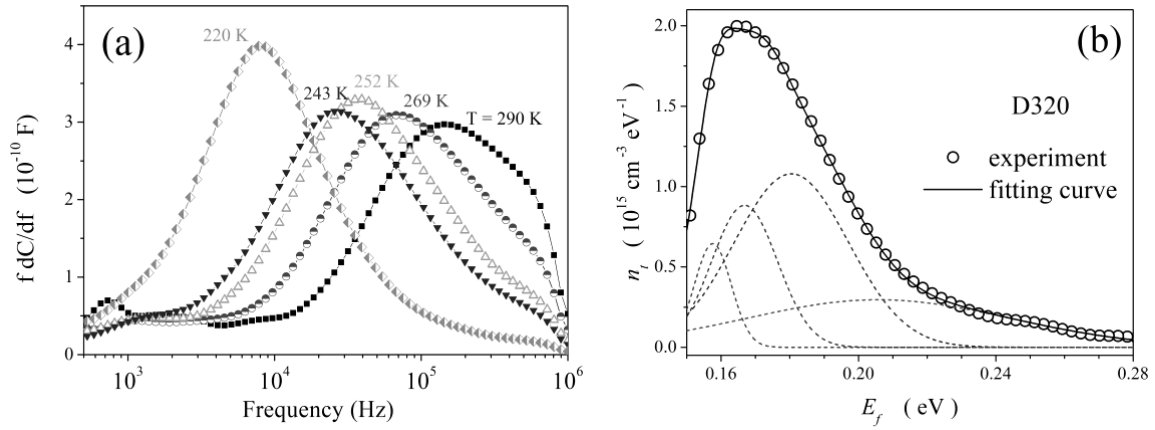


Fig. 1. (a) Frequency dependence of the capacitance derivative of the PEDOT:PSS/Si structure at different temperatures. (b) Representative Gaussian decomposition of the tDOS energy spectrum. Symbols denote experimental data, dashed lines correspond to individual defect contributions, and the solid line represents the cumulative fit.

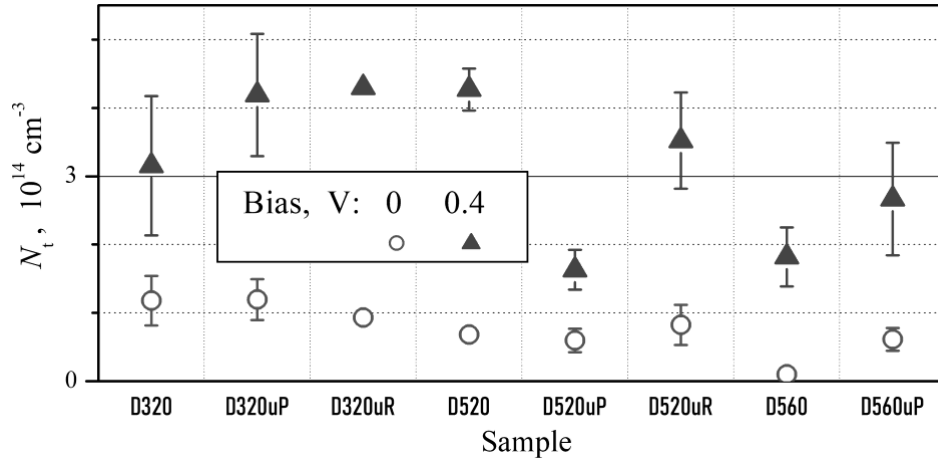


Fig. 2. Total tDOS determined at different bias voltages for samples fabricated under various processing conditions. In the sample notation, the first digit denotes the spin-coating speed, where 3 and 5 correspond to 3000 and 5000 rpm, respectively, and the following two digits indicate the room-temperature relaxation time in minutes. Suffixes specify ultrasonic treatment, with “uP” denoting primary waves and “uR” denoting radial waves.

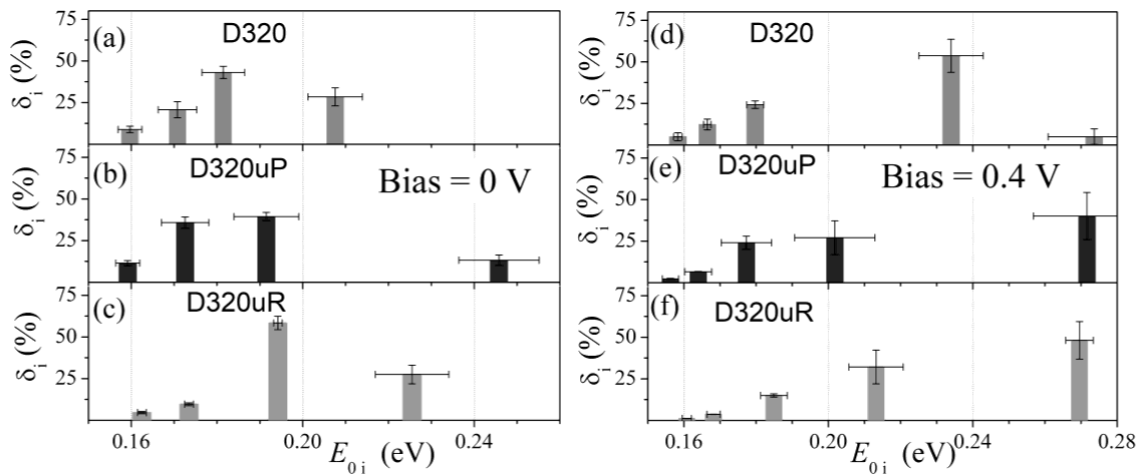


Fig. 3. Contributions of different defect types as a function of energy at bias voltages of 0 V (a–c) and 0.4 V (d–f). Data are shown for samples fabricated at a spin-coating speed of 3000 rpm with a 20 min room-temperature relaxation time: (a, d) without ultrasound treatment; (b, e) under primary-wave excitation; and (c, f) under radial-wave excitation.